전산화학 (2014년 1학기) **1차 시험 해답**

(총 6 문제, 60 점)

1. **(총 15점, 각 5점)**

(a) $-\frac{\sqrt{3}}{2} - \frac{i}{2} = \big(\cos(7\pi/6) + i\sin(7\pi/6)\big) = e^{i7\pi/6}$ 이므로

$$\begin{aligned} z = \left(-\frac{\sqrt{3}}{2} - \frac{i}{2}\right)^{27} &= \left(e^{i7\pi/6}\right)^{27} \\ &= e^{i(7\pi/6)\times 27} \\ &= e^{i189\pi/6} \\ &= e^{i(180\pi/6+9\pi/6} \\ &= e^{i(30\pi+3\pi/2)} \\ &= e^{i3\pi/2} \\ &= \cos\left(\frac{3\pi}{2}\right) + i\sin\left(\frac{3\pi}{2}\right) \\ &= -i \end{aligned}$$

$\therefore \mathrm{Re}(z) = 0, \qquad \mathrm{Im}(z) = -1.$

(b)

$$\begin{aligned} z = e^2 \times e^{(-3\pi/4)i} &= e^2\left\{\cos\left(-\frac{3\pi}{4}\right) + i\sin\left(-\frac{3\pi}{4}\right)\right\} \\ &= e^2\left(-\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}\right) \\ &= -\frac{\sqrt{2}}{2}e^2 - i\frac{\sqrt{2}}{2}e^2 \end{aligned}$$

$\therefore \mathrm{Re}(z) = -\frac{\sqrt{2}}{2}e^2, \qquad \mathrm{Im}(z) = -\frac{\sqrt{2}}{2}e^2.$

(c)

$$\begin{aligned} \sqrt[3]{-1} &= \big\{\cos\big(2n\pi+\pi\big) + \sin\big(2n\pi+\pi\big)\big\}^{1/3} \qquad (n = 0, 1, 2, \cdots) \\ &= \cos\left(\frac{2n\pi+\pi}{3}\right) + \sin\left(\frac{2n\pi+\pi}{3}\right) \end{aligned} \tag{1}$$

i. $n = 0$ 인 경우, (1) 식은

$$\sqrt[3]{-1} = \cos\left(\frac{2n\pi+\pi}{3}\right) + \sin\left(\frac{2n\pi+\pi}{3}\right) = \cos\left(\frac{\pi}{3}\right) + \sin\left(\frac{\pi}{3}\right) = \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

ii. $n = 1$ 인 경우, (1) 식은

$$\sqrt[3]{-1} = \cos\left(\frac{2n\pi+\pi}{3}\right) + \sin\left(\frac{2n\pi+\pi}{3}\right) = \cos\big(\pi\big) + \sin\big(\pi\big) = -1$$

iii. $n = 2$ 인 경우, (1) 식은

$$\sqrt[3]{-1} = \cos\left(\frac{2n\pi+\pi}{3}\right) + \sin\left(\frac{2n\pi+\pi}{3}\right) = \cos\left(\frac{5\pi}{3}\right) + \sin\left(\frac{5\pi}{3}\right) = \frac{1}{2} - \frac{\sqrt{3}}{2}i$$

한편,

$$\begin{aligned} \sqrt[3]{i} &= \left\{\cos\left(2n\pi+\frac{\pi}{2}\right) + \sin\left(2n\pi+\frac{\pi}{2}\right)\right\}^{1/3} \qquad (n = 0, 1, 2, \cdots) \\ &= \cos\left(\frac{2n\pi}{3}+\frac{\pi}{6}\right) + \sin\left(\frac{2n\pi}{3}+\frac{\pi}{6}\right) \end{aligned} \tag{2}$$

i. $n=0$ 인 경우, (3) 식은

$$\sqrt[3]{i}=\cos\left(\frac{2n\pi}{3}+\frac{\pi}{6}\right)+\sin\left(\frac{2n\pi}{3}+\frac{\pi}{6}\right)=\cos\left(\frac{\pi}{6}\right)+\sin\left(\frac{\pi}{6}\right)=\frac{\sqrt{3}}{2}+\frac{1}{2}i$$

ii. $n=1$ 인 경우, (3) 식은

$$\sqrt[3]{i}=\cos\left(\frac{2n\pi}{3}+\frac{\pi}{6}\right)+\sin\left(\frac{2n\pi}{3}+\frac{\pi}{6}\right)=\cos\left(\frac{5\pi}{6}\right)+\sin\left(\frac{5\pi}{6}\right)=-\frac{\sqrt{3}}{2}+\frac{1}{2}i$$

iii. $n=2$ 인 경우, (3) 식은

$$\sqrt[3]{i}=\cos\left(\frac{4n\pi}{3}+\frac{\pi}{6}\right)+\sin\left(\frac{4n\pi}{3}+\frac{\pi}{6}\right)=\cos\left(\frac{3\pi}{2}\right)+\sin\left(\frac{3\pi}{2}\right)=-i$$

그러므로

$$\begin{aligned}
z=\sqrt[3]{-1}+\sqrt[3]{i} &= \frac{1}{2}+\frac{\sqrt{3}}{2}i+\frac{\sqrt{3}}{2}+\frac{1}{2}i=\left(\frac{1}{2}+\frac{\sqrt{3}}{2}\right)+\left(\frac{\sqrt{3}}{2}+\frac{1}{2}\right)i, \qquad \text{또는}\\
&= \frac{1}{2}+\frac{\sqrt{3}}{2}i-\frac{\sqrt{3}}{2}+\frac{1}{2}i=\left(\frac{1}{2}-\frac{\sqrt{3}}{2}\right)+\left(\frac{\sqrt{3}}{2}+\frac{1}{2}\right)i, \qquad \text{또는}\\
&= \frac{1}{2}+\frac{\sqrt{3}}{2}i-i=\frac{1}{2}+\left(\frac{\sqrt{3}}{2}-1\right)i, \qquad \text{또는}\\
&= -1+\frac{\sqrt{3}}{2}+\frac{1}{2}i=\left(-1+\frac{\sqrt{3}}{2}\right)+\frac{1}{2}i, \qquad \text{또는}\\
&= -1-\frac{\sqrt{3}}{2}+\frac{1}{2}i=\left(-1-\frac{\sqrt{3}}{2}\right)+\frac{1}{2}i, \qquad \text{또는}\\
&= -1-i, \qquad \text{또는}\\
&= -\frac{1}{2}-\frac{\sqrt{3}}{2}i+\frac{\sqrt{3}}{2}+\frac{1}{2}i=\left(-\frac{1}{2}+\frac{\sqrt{3}}{2}\right)+\left(-\frac{\sqrt{3}}{2}+\frac{1}{2}\right)i, \qquad \text{또는}\\
&= -\frac{1}{2}-\frac{\sqrt{3}}{2}i-\frac{\sqrt{3}}{2}+\frac{1}{2}i=\left(-\frac{1}{2}-\frac{\sqrt{3}}{2}\right)+\left(-\frac{\sqrt{3}}{2}+\frac{1}{2}\right)i, \qquad \text{또는}\\
&= -\frac{1}{2}-\frac{\sqrt{3}}{2}i-i=-\frac{1}{2}-\left(\frac{\sqrt{3}}{2}+1\right)i
\end{aligned}$$

그러므로

$\mathrm{Re}(z)=\frac{1}{2}+\frac{\sqrt{3}}{2}$, $\mathrm{Im}(z)=\frac{\sqrt{3}}{2}+\frac{1}{2}$, 또는 $\mathrm{Re}(z)=\frac{1}{2}-\frac{\sqrt{3}}{2}$, $\mathrm{Im}(z)=\frac{\sqrt{3}}{2}+\frac{1}{2}$, 또는
$\mathrm{Re}(z)=\frac{1}{2}$, $\mathrm{Im}(z)=\frac{\sqrt{3}}{2}-1$, 또는 $\mathrm{Re}(z)=-1+\frac{\sqrt{3}}{2}$, $\mathrm{Im}(z)=\frac{1}{2}$, 또는
$\mathrm{Re}(z)=-1-\frac{\sqrt{3}}{2}$, $\mathrm{Im}(z)=\frac{1}{2}$, 또는 $\mathrm{Re}(z)=-1$, $\mathrm{Im}(z)=-1$, 또는
$\mathrm{Re}(z)=-\frac{1}{2}+\frac{\sqrt{3}}{2}$, $\mathrm{Im}(z)=-\frac{\sqrt{3}}{2}+\frac{1}{2}$, 또는 $\mathrm{Re}(z)=-\frac{1}{2}-\frac{\sqrt{3}}{2}$, $\mathrm{Im}(z)=-\frac{\sqrt{3}}{2}+\frac{1}{2}$,
또는 $\mathrm{Re}(z)=-\frac{1}{2}$, $\mathrm{Im}(z)=-\left(\frac{\sqrt{3}}{2}+\frac{1}{2}\right)$.

2. **(총 10점, 각 5점)**

(a) $z=a+ib$ 라면,

$$\mathrm{Re}(z(i+1))+zz^*=\mathrm{Re}(a-b+(a+b)i)+a^2+b^2=a-b+a^2+b^2=0 \quad (3)$$

$$\mathrm{Im}(z(i+1))=\mathrm{Im}(a-b+(a+b)i)=a+b=0\Rightarrow b=-a \quad (4)$$

(4) 식을 (3) 식에 대입하면

$$2a^2+2a=0\Rightarrow a(a+1)=0 \qquad a=0 \qquad \text{또는} \qquad a=-1 \quad (5)$$

i. $a = 0$ 인 경우 (4)로부터 $b = 0$, 그러므로 $z = 0$.

ii. $a = -1$ 인 경우 (4)로부터 $b = 1$, 그러므로 $z = -1 + i$.

(b) $z = a + ib$라면,

$$\begin{aligned} -3 = \mathrm{Re}(z^2) + i\mathrm{Im}(z^*(1+2i)) &= \mathrm{Re}((a+ib)^2) + i\mathrm{Im}((a-ib)(1+2i)) \\ &= \mathrm{Re}(a^2 - b^2 + 2abi) + i\mathrm{Im}(a + 2b + i(2a-b)) \\ &= a^2 - b^2 + i(2a-b) \qquad (6) \end{aligned}$$

복소수 상등 정리에 의해 (6) 식으로부터 $a^2 - b^2 = -3$, 그리고 $2a - b = 0$. 두 식을 연립하면,

i. $a = 1$ 인 경우 $b = 2$, 그러므로 $z = 1 + 2i$.

ii. $a = -1$ 인 경우 $b = -2$, 그러므로 $z = -1 - 2i$.

3. **(총 10점, 각 5점)**

(a)

$$\begin{aligned} x &= r\sin\theta\cos\phi = 2 \times \sin\left(\frac{\pi}{4}\right) \times \cos\left(-\frac{3\pi}{2}\right) = 2 \times \frac{\sqrt{2}}{2} \times (0) = 0 \\ y &= r\sin\theta\sin\phi = 2 \times \sin\left(\frac{\pi}{4}\right) \times \sin\left(-\frac{3\pi}{2}\right) = 2 \times \frac{\sqrt{2}}{2} \times (-1) = -\sqrt{2} \\ z &= r\cos\theta = 2 \times \cos\left(\frac{\pi}{4}\right) = 2 \times \frac{\sqrt{2}}{2} = \sqrt{2} \end{aligned}$$

(b)

$$\begin{aligned} r &= \sqrt{x^2+y^2+z^2} = \sqrt{(-1)^2 + 4^2 + (-2)^2} = \sqrt{21} \\ \theta &= \arccos\left(\frac{z}{r}\right) = \arccos\left(\frac{-2}{\sqrt{21}}\right) = \arccos\left(-\frac{2}{\sqrt{21}}\right) \\ \phi &= \arctan\left(\frac{y}{x}\right) = \arctan\frac{4}{-1} = \arctan(-4) \end{aligned}$$

4. **(총 15점, 각 5점)**

(a)

$$\begin{aligned} d(r,\theta,\phi) &= \sin\theta\cos\theta\cos\phi \\ &= \frac{(r\sin\theta\cos\phi)(r\cos\theta)}{r^2} \\ d(x,y,z) &= \frac{xz}{x^2+y^2+z^2} \end{aligned}$$

(b)

$$\begin{aligned} d(r,\theta,\phi) &= \sin^2\theta\cos 2\phi \\ &= \sin^2\theta(\cos^2\phi - \sin^2\phi) \\ &= \sin^2\theta\cos^2\phi - \sin^2\theta\sin^2\phi \\ &= \frac{r^2\sin^2\theta\cos^2\phi - r^2\sin^2\theta\sin^2\phi}{r^2} \\ &= \frac{(r\sin\theta\cos\phi)^2 - (r\sin\theta\sin\phi)^2}{r^2} \\ d(x,y,z) &= \frac{x^2-y^2}{x^2+y^2+z^2} \end{aligned}$$

(c)

$$\begin{aligned} d(r,\theta,\phi) &= \sin^2\theta \sin 2\phi \\ &= \sin^2\theta(2\sin\phi\cos\phi) \\ &= 2\sin\theta\sin\phi\sin\theta\cos\phi \\ &= \frac{2(r\sin\theta\sin\phi)(r\sin\theta\cos\phi)}{r^2} \\ d(x,y,z) &= \frac{2yx}{x^2+y^2+z^2} \end{aligned}$$

5. **(총 10점, 각 5점)**

(a) $m = n$인 경우,

$$\int_0^{2\pi} d\phi \Phi_m(\phi)\Phi_n^*(\phi) = \int_0^{2\pi} d\phi \frac{1}{\sqrt{2\pi}} e^{im\phi} \frac{1}{\sqrt{2\pi}} e^{-im\phi} = \frac{1}{2\pi}\int_0^{2\pi} d\phi = \frac{1}{2\pi}(2\pi) = 1.$$

(b) $m \neq n$인 경우,

$$\begin{aligned} \int_0^{2\pi} d\phi \Phi_m(\phi)\Phi_n(\phi) &= \int_0^{2\pi} d\phi \frac{1}{\sqrt{2\pi}} e^{im\phi} \frac{1}{\sqrt{2\pi}} e^{-in\phi} \\ &= \frac{1}{2\pi}\int_0^{2\pi} d\phi e^{i(m-n)\phi} \\ &= \frac{1}{2\pi}\int_0^{2\pi} d\phi \{\cos(m-n)\phi + i\sin(m-n)\phi\} \\ &= \frac{1}{2\pi}\left[\frac{1}{m-n}\sin(m-n)\phi - i\frac{1}{m-n}\cos(m-n)\phi\right]_0^{2\pi} \\ &= \frac{1}{2\pi}\frac{1}{m-n}\Big\{\sin((m-n)2\pi) - \sin(0) - i(\cos((m-n)2\pi) - \cos(0))\Big\} \\ &= 0. \qquad (m-n\text{은 0이 아닌 정수이므로}) \end{aligned}$$